

**MOTOROLA**

MC68020 MC68EC020

MICROPROCESSORS USER'S MANUAL

First Edition

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PREFACE

The *M68020 User's Manual* describes the capabilities, operation, and programming of the MC68020 32-bit, second-generation, enhanced microprocessor and the MC68EC020 32-bit, second-generation, enhanced embedded microprocessor.

Throughout this manual, "MC68020/EC020" is used when information applies to both the MC68020 and the MC68EC020. "MC68020" and "MC68EC020" are used when information applies only to the MC68020 or MC68EC020, respectively.

For detailed information on the MC68020 and MC68EC020 instruction set, refer to M68000PM/AD, *M68000 Family Programmer's Reference Manual*.

This manual consists of the following sections:

Section 1	Introduction
Section 2	Processing States
Section 3	Signal Description
Section 4	On-Chip Cache Memory
Section 5	Bus Operation
Section 6	Exception Processing
Section 7	Coprocessor Interface Description
Section 8	Instruction Execution Timing
Section 9	Applications Information
Section 10	Electrical Characteristics
Section 11	Ordering Information and Mechanical Data
Appendix A	Interfacing an MC68EC020 to a DMA Device That Supports a Three-Wire Bus Arbitration Protocol

NOTE

In this manual, *assert* and *negate* are used to specify forcing a signal to a particular state. In particular, *assertion* and *assert* refer to a signal that is active or true; *negation* and *negate* indicate a signal that is inactive or false. These terms are used independently of the voltage level (high or low) that they represent.